

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Winter Examination–2022

Course: B. Tech. Branch :Information Technology Semester :III
Subject Code &Name: BTITC305, Data Structures and Applications
Max Marks: 60 Date:17/03/2023 Duration: 3 Hr.

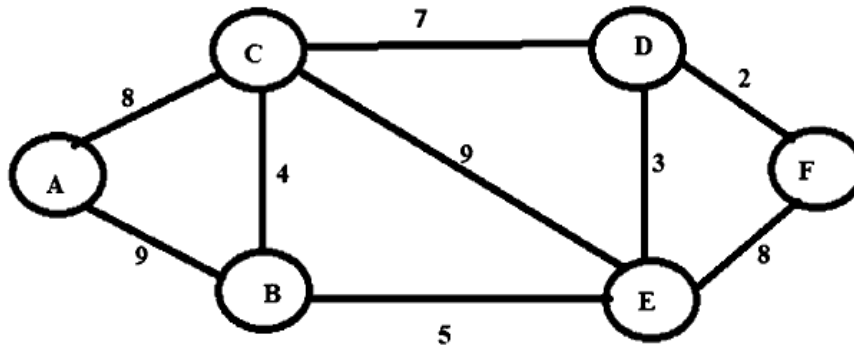
Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve Any Two of the following.		12
A) Illustrate the classification of Data Structures with an example of each class.	Apply CO1	6
B) Describe Time and Space complexity related to an algorithm.	Understand CO1	6
C) Write an algorithm for the following statements.	Create	6
i) To check whether a given number is prime	CO1	
ii) To find Factorial of a number		
Q.2 Solve Any Two of the following.		12
A) Evaluate the given postfix expression using stack by assuming values given below:	Evaluate CO2	6
P=2, Q=4, R=6, S=8, T=12, U=12, V=2, W=6, X=6		
PQ+RS*T/-UV*WX+/+		
B) Draw and explain in detail the circular queue with an example.	Understand CO2	6
C) Design a procedure which finds number NUM of times a given ITEM occurs in a linked list.	Create CO2	6
Q. 3 Solve Any Two of the following.		12
A) Explain preorder traversal of a binary tree with an example.	Understand CO3	6
B) Construct a tree for given expression.	Create	6
$(10a^2+5b)^5+(3a-4b^5)^3$	CO3	
C) What is a binary tree? Write any three types of binary tree with an example.	Knowledge CO3	6

Q.4 Solve Any Two of the following. **12**

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|--|-------------------|---|
| A) Explain Breadth First Search (BFS) technique along with an algorithm. | Understand
CO4 | 6 |
| B) Find the minimum spanning tree using Kruskal's algorithm for the given graph. | Evaluate
CO4 | 6 |



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|---|-------------------|---|
| C) How the graphs are represented in a memory? Explain with an example. | Understand
CO4 | 6 |
|---|-------------------|---|

Q. 5 Solve Any Two of the following. **12**

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|--|-----------------------------|---|
| A) Write an algorithm for linear search in an array. | Understand
CO5 | 6 |
| B) Sort the following list using Merge Sort. Write its complexity.
12, 8, 2, 6, 25, 5, 4, 9 | Apply
CO5 | 6 |
| C) Write and explain Quick Sort algorithm to sort a given array. | Create
Understand
CO5 | 6 |

*** End ***